Supplementary information

Target-correlation maps are library-conditional but recoverable from a few hundred ligands in two large Vina panels

Adeliya Leleytner, Victor Safronov, Maxim Fedorov

S1. Scope, data and estimand contracts

This Supplement follows the audit stages in the main Results: how concentrated the two large Vina matrices are; what remains after removal of additive ligand and target effects; what target-specific ligand-feature fits can and cannot identify, and how the map transports across chemical domains; whether the target-pair geometry agrees with experimental co-response; how that agreement depends on protein and target support; and whether the transformation improves within-ligand target ranking. The two score matrices are independently sourced rather than statistically independent samples. Their exact chemical and target overlaps are reported in Table S23.

For a ligand-by-target score matrix X, "column-standardized" denotes target-wise mean-zero, unit-variance scores. "Two-way residual" denotes the residual from the additive model

$$X_{ij} = \mu + a_i + b_j + \Gamma_{ij},$$

followed by target-wise standardization when a correlation spectrum or scaled residual score is required. Correlation spectra therefore give each target equal marginal variance; covariance spectra retain target-specific variance. These are different estimands, and both are reported in Table S3. A target-pair geometry is the vector of unique off-diagonal correlations of the specified transformed matrix.

The external geometry analysis used the same 20 kinases and the same 190 target pairs in DAVIS, PKIS2, PKIS1 and KiRHub. Experimental matrices were represented both before and after two-way centring, giving the complete 2×2 docking-by-experiment comparison in Table S7. The fixed 15-pair endpoint was defined as pairs in the top decile of centred experimental correlation in at least two of DAVIS, PKIS2 and PKIS1; the complete locked set is given in Table S12. KiRHub was inspected only after this endpoint was fixed, and no KiRHub result was used to alter it. This ordering was not preregistered. Because the endpoint is defined from centred experimental geometry, raw-versus-centred retrieval gaps against it are conditional on the endpoint construction and cannot be interpreted as a causal or transformation-neutral benefit of centring. Across the declared 5–25% threshold family, the mean centred-minus-raw gain was unresolved (target-label QAP $p = 0.061$ for AUROC and $p = 0.076$ for average precision). For the three panels with released structures, identity and scaffold intersections are reported in Table S24. KiRHub provides compound names without a frozen molecular-structure mapping, so only exact normalized-name overlap with DAVIS was auditable.

The operational ChEMBL benchmark is a distinct estimand. It scores only experimentally co-observed target pairs within each ligand and does not impute unmeasured activities or test retrieval of an unmeasured target. The broad primary benchmark matches ChEMBL to DOCKSTRING and contains 6,557 retained ligands, 31 targets, 18,020 observed cells and 25,214 non-tied within-ligand target comparisons among 6,480 evaluated ligands. Chemical cluster intervals condition on the fixed target labels; delete-one-target jackknife intervals are reported separately as panel-composition sensitivities. The endpoint-restricted, less heterogeneous human binding Ki/Kd arm and the separate Ki, Kd, IC50 and EC50 arms alter the experimental endpoint contract but retain the externally fitted docking transformation.

The earlier Docking-44–ChEMBL benchmark contains 137 ligands, 38 targets, 691 observed cells and 2,522 non-tied comparisons. It is retained only as a legacy sparse sensitivity and does not contribute the primary operational estimate.

Two dense ligand-wise checks use exact molecular-structure matches to released DOCKSTRING profiles. DAVIS and PKIS2 contain 59 and 154 matched ligands across 21 shared kinases; the extra target relative to the fixed-20 geometry analysis is PTK2/FAK. These analyses evaluate target order for observed ligand–target cells, not the correlation structure across a different chemical library and not retrieval of an unmeasured target.

S2. Spectral concentration and residual dependence

For a P-target correlation matrix C with eigenvalues λ_k, the participation-ratio (PR) effective dimension is

$$r_{\mathrm{PR}} = \frac{(\sum_k \lambda_k)^2}{\sum_k \lambda_k^2} = \frac{P}{1 + (P-1)\overline{r_{ij}^2}}.$$

The final equality holds for a correlation matrix. PR is therefore an interpretable re-expression of mean-squared inter-target correlation, not a new estimator or an algebraic rank. Complementary scalar summaries are collected in Table S1; the full spectrum (Fig. S1), signed correlation distributions and heatmaps (Fig. S3) retain information that PR discards.

Docking-44 increased from PR 1.834 to 9.303 after two-way centring; DOCKSTRING-58 increased from 2.266 to 18.206. The last residual eigenvalue is zero by construction because every residual row sums to zero. The same qualitative increase held for covariance spectra, although the residual values were lower than for correlation spectra (Table S3; Fig. S4c,d). Thus the conclusion is not created by unit-variance scaling, while its magnitude depends on whether target variances are retained.

The concentration was also observed when the tested functions rescored the same Vina-selected poses. On a fixed 2,000-ligand by 9-target block of released DOCKSTRING poses, with only the scorer changed, 5 of 6 non-Vina functions met all four stated criteria; smina's re-implementation of Vina reproduced the released scores at Pearson ≥ 0.975 per target (Table S15). PLECscore is the least concentrated function tested (raw PC1 41.4%, raw PR 4.59) and simultaneously the one whose leading axis is closest to uniform (cosine 0.986); its PR also rises under centring.

Three qualifications belong with that table. First, the four criteria are a weak bar: a one-factor surface carrying no target identity satisfies all of them across a wide noise range, and Perron–Frobenius makes the near-uniform leading eigenvector close to automatic for an all-positive correlation matrix, so passing the set evidences a shared positive ligand axis and nothing finer. The discriminating quantity is instead the centred PR, which reaches only 5.12–7.47 here against roughly 7.9–8.0 for that null, so every scorer retains genuine residual cross-target correlation; standardizing columns before centring moves centred PR by at most 0.4, so the rise is not a column-scale artefact. Second, two margins sit inside bootstrap noise and are reported as point-estimate calls rather than decisions: Vina's own uniform cosine is 0.95007 against a 0.95 threshold, and PLECscore's raw PR exceeds the 4.5 threshold at the point estimate with an interval that spans it, whereas its PC1 failure is decisive in every replicate. Third, the six non-Vina functions are four independent lineages: RF-Score v1, v2 and v3 share one PDBbind-2016 training table and correlate at 0.89–0.94 at cell level, v3 additionally consumes Vina terms as features, and all the machine-learned scorers are trained on crystal complexes but applied to docked poses. Rescoring a retained pose is in any case an upper bound on the agreement a full re-docking run would show.

Only 1.466% of Docking-44 cells were missing, but they occurred in 3,354 ligands; target 4mqs accounted for 3,352 missing cells. Mean, median and observed-cell additive least-squares imputations gave residual PR values of 9.303, 9.528 and 9.663. Complete-case deletion gave 13.970 on 9,297 chemically selected ligands, and deletion of 4mqs gave 9.433 under mean imputation (Table S2; Fig. S4a,b). Complete rows had median molecular weight 249 Da and 17 heavy atoms, whereas incomplete rows had medians 399 Da and 28 heavy atoms. Complete-case analysis therefore changes chemical support rather than providing a like-for-like missing-value correction.

DOCKSTRING v1 contained 260,155 rows, of which 95 had at least one of 339 missing scores. Listwise deletion therefore removed only 0.0365% of molecules, leaving the 260,060-row primary matrix. Repeating the analysis on all released rows after target-mean imputation gave raw and residual PR values of 2.267 and 18.214; target-median imputation gave 2.267 and 18.214. These values were effectively unchanged from 2.266 and 18.206 on complete rows, unlike the chemically selective Docking-44 complete-case analysis.

Target composition was examined by leave-one-target-out deletion (Fig. S2; Table S5), random and family-stratified target subsampling (Table S6), and chemical- support resampling. Five independently seeded 15,000-molecule DOCKSTRING supports gave raw PR 2.252–2.294 and residual PR 18.161–18.322 (Fig. S6). Their intervals resample entire Bemis–Murcko scaffold groups. These ranges describe the observed panel and source library; they are not confidence intervals for all proteins or all chemical space.

Four matched nulls addressed different mechanical explanations. Independent column permutation supplied the transformation-matched parallel-analysis null. On a common 12,000-row support, the descriptive numbers above the rank-wise 95th-percentile null were 3 versus 7 for Docking-44 and 4 versus 11 for DOCKSTRING-58 before versus after centring. A studentized maximum-deviation envelope controlling family-wise error across eigenvalue ranks retained 3 versus 7 and 4 versus 10 modes. Mode counts are sample-size-dependent descriptive summaries and are not used as the primary measure of practical dimension.

The fitted additive Gaussian null retained estimated ligand and target effects and target-specific residual variances. The empirical column-permutation null retained every target's non-Gaussian residual marginal. The row-norm-preserving null drew an independent random direction in the target-zero-sum subspace for each ligand and rescaled it to the observed residual row norm. Observed residual PR values were far below all three matched nulls (Table S4); under the row-norm null they were 9.295 versus a median of 42.628 and 18.158 versus 56.612 ($p = 1/501 = 0.002$ for each). No single null preserves all empirical structure. The row-norm null medians lie near the residual algebraic ceilings of 43 and 57, so this comparison establishes structured dependence beyond the matched zero-sum and row-norm constraints rather than its origin or biological validity; agreement across differently matched nulls is the relevant result.

S3. Ligand-feature interpretation and rank-matched controls

Raw PC1 loadings were oriented to have positive sum. Their cosine similarities with the uniform target vector were 0.970 and 0.969 (Fig. S5a,b), and the corresponding ligand scores correlated 0.998 and 0.997 with the mean column-standardized score. This identifies the leading component as nearly, but not exactly, the uniform ligand-mean direction.

For each target, ordinary least squares fitted

$$\text{Vina score} = \beta_0 + \beta_{\mathrm{MW}}(\mathrm{MW}/100) + \beta_{\mathrm{rot}} N_{\mathrm{rot}}.$$

Docking-44 used all 12,651 mean-imputed rows. DOCKSTRING-58 used a fixed score-blind random sample of 15,000 from 260,060 ligands (seed 71), because descriptor generation for the full matrix was not required for this interpretation. Median conditional molecular-weight slopes were -0.350 and -0.778 kcal/mol per 100 Da, while median RDKit rotatable-bond slopes were +0.100 and +0.153 kcal/mol per bond. The flexibility slope was positive for 44/44 and 57/58 targets (Fig. S5c,d).

Two-way centring converts a descriptor's target-specific raw slope vector into its mean-centred slope vector. The molecular-weight residual-correlation vector aligned with the residual PC1 loading vector at cosines 0.963 and 0.929; absolute correlations between molecular weight and residual PC1 ligand scores were 0.658 and 0.734 (Fig. S5e,f). Removing 4mqs from Docking-44 left the two values at 0.960 and 0.647. These results support target-dependent physicochemical

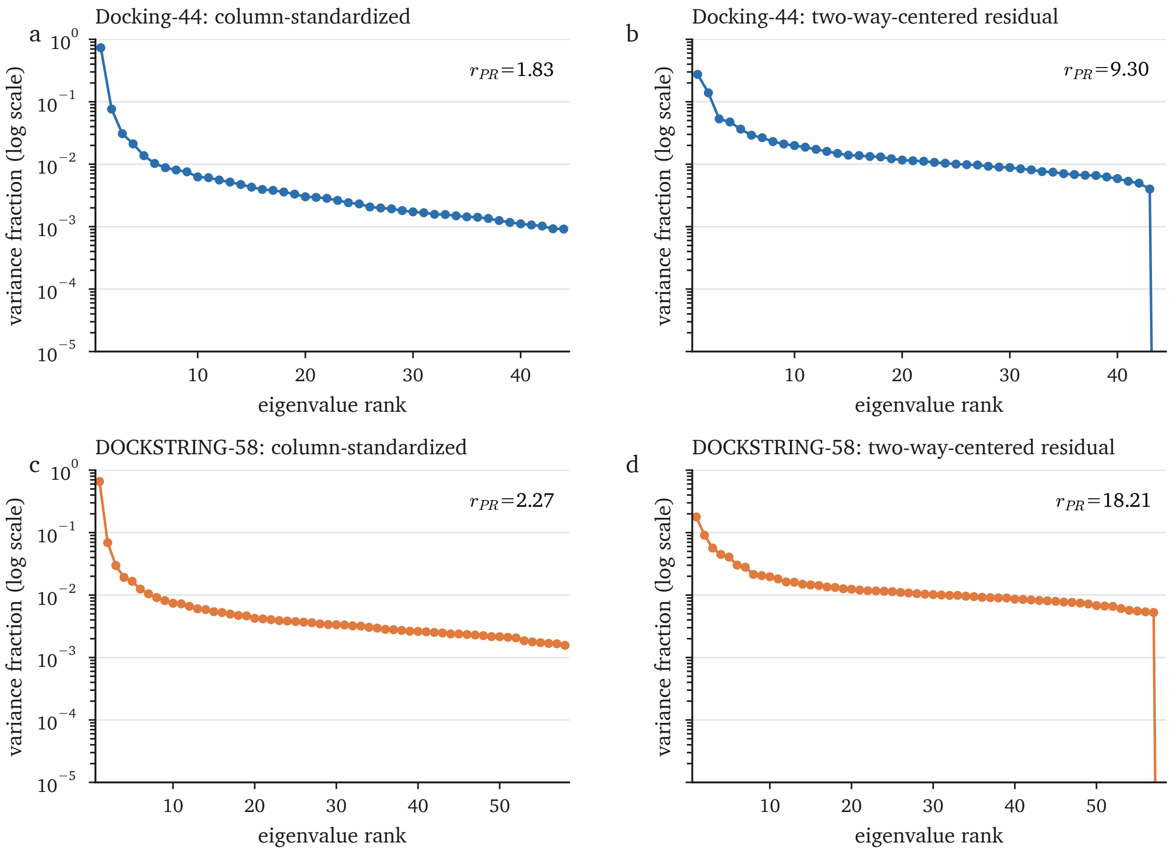

Figure S1: Full correlation-spectrum scree plots. a,b, Docking-44 column-standardized and two-way-residual spectra. **c,d**, DOCKSTRING-58 column-standardized and two-way-residual spectra. Eigenvalues are divided by the number of targets so each spectrum sums to one. The ordinate is logarithmic. Two-way centring imposes one exact zero residual eigenvalue.

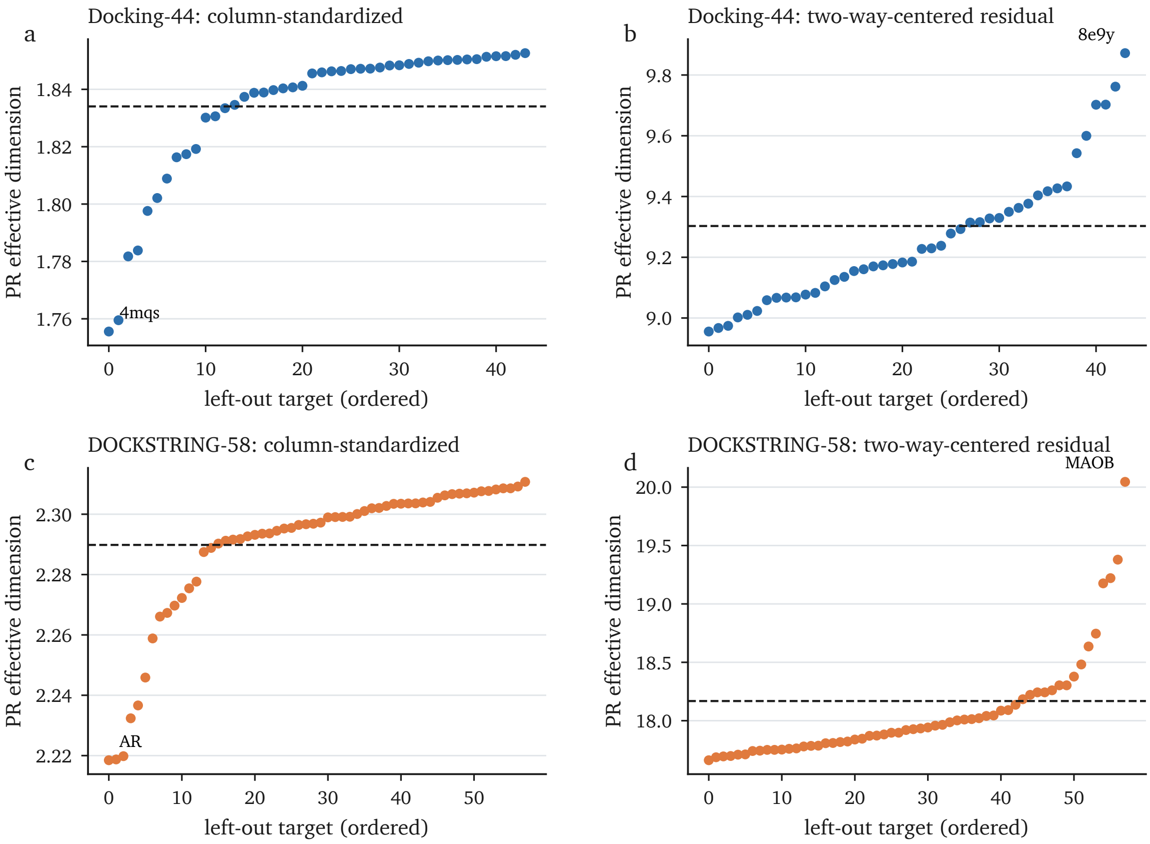

Figure S2: Leave-one-target-out spectral composition sensitivity. Each point is the PR effective dimension after deleting one target; targets are ordered by the resulting value. Dashed lines mark the complete-panel estimate on the same ligand support. Labels identify the most influential deletion. These are fixed-panel sensitivity ranges rather than intervals for a target superpopulation.

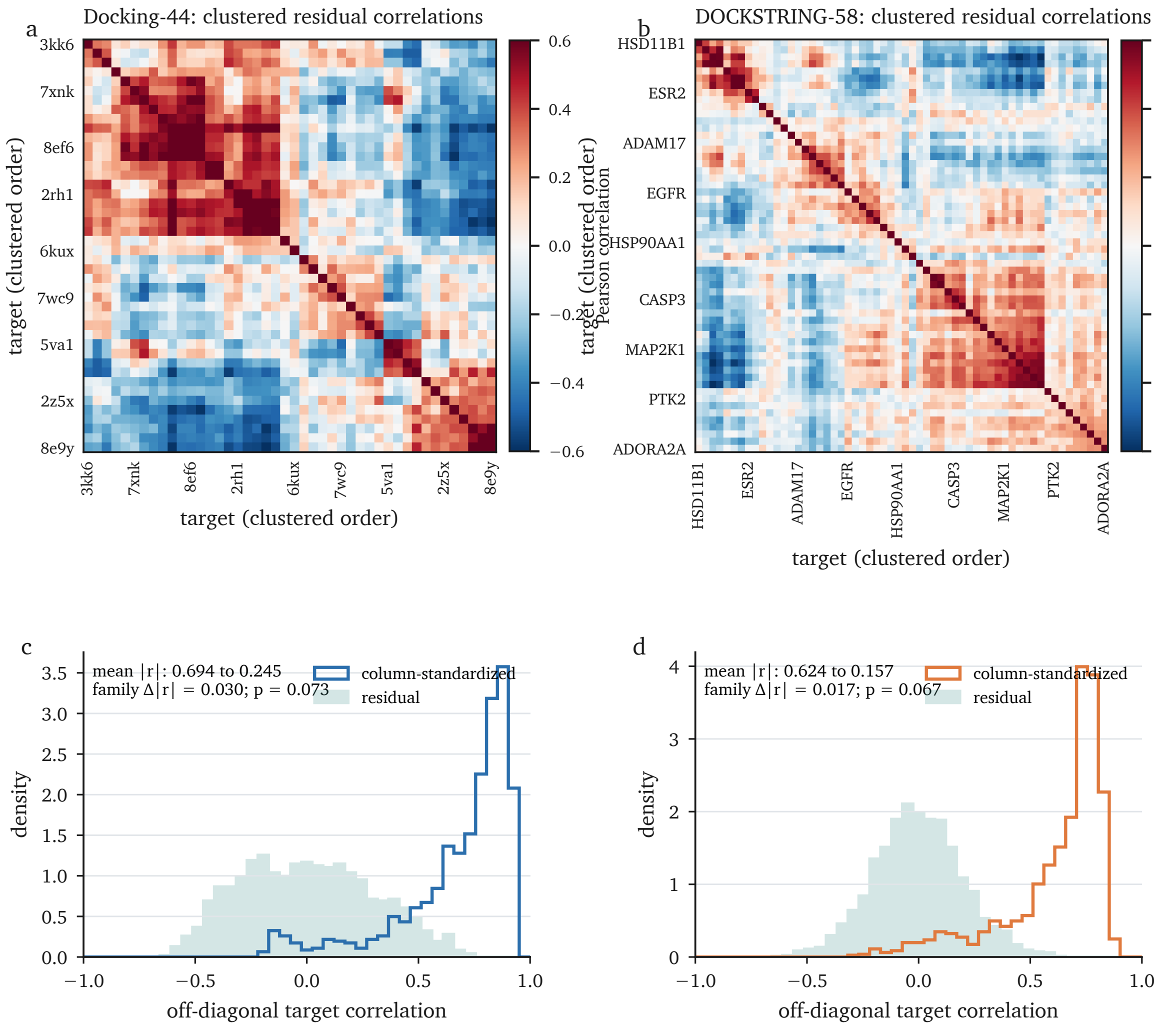

Figure S3: Centring exposes signed residual target dependence. a,b, Two-way-residual target correlations ordered by average-linkage clustering of $1 - r$. A subset of labels is printed for legibility; complete matrices and target order are machine-readable. **c,d**, Distributions of off-diagonal correlations before and after centring, on fixed 12,000-ligand Docking-44 and 15,000-ligand DOCKSTRING supports. Mean absolute correlation fell from 0.694 to 0.245 and from 0.624 to 0.157, respectively. These support-specific values differ slightly from the full-support summaries in Table S1. Broad-family label permutations did not resolve the small within-minus-between-family differences ($p = 0.073$ and 0.067).

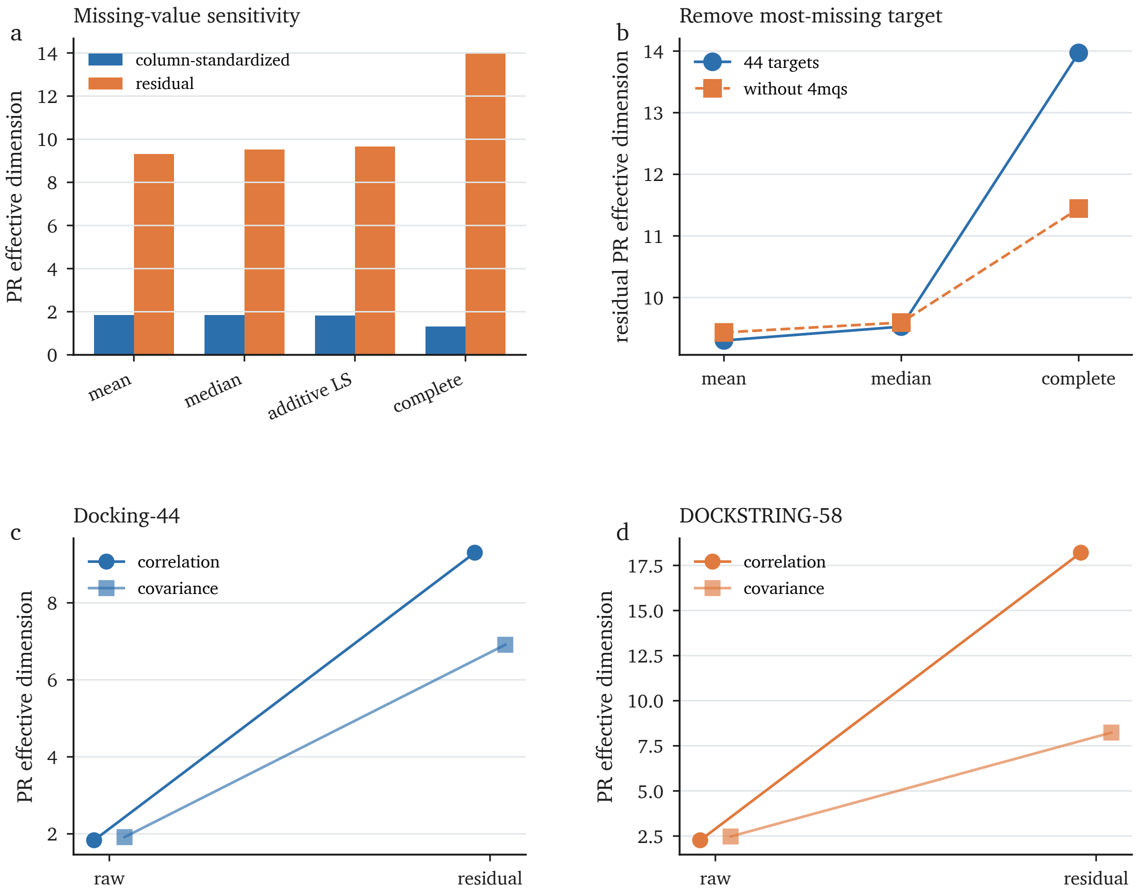

Figure S4: Missing-value and spectrum-estimand sensitivities. **a**, Docking-44 PR under four missing-score strategies. **b**, Residual PR before and after deletion of 4mqs, the target with most missing scores. **c,d**, Correlation- and covariance-spectrum PR before and after two-way centring. Complete-case values are not directly comparable with full-support imputations because the chemical support changes.

slopes as one source of the leading residual mode. RDKit rotatable-bond count is a proxy, not Vina's internal torsion term, so this analysis is not a decomposition of the Vina score.

We next fitted seven ligand descriptors—heavy-atom count, molecular weight, Labute ASA, TPSA, cLogP, RDKit rotatable-bond count and ring count—to each residual target column in five chemical-group-held-out folds. Every target offset, residual target scale, descriptor scale and coefficient was estimated on the training fold; pooled errors were strictly out-of-fold. On the full matrices, subtracting this prediction reduced mean squared target correlation by 46.9% and 53.6%, while PR increased from 9.294 to 14.746 and from 17.923 to 28.466 (Table S14). These are predictive contrasts, not causal or feature-specific attributions.

The fixed common-20 kinase support allowed rank-matched controls using the same folds and 259,579 DOCKSTRING ligands after excluding structural overlaps with DAVIS, PKIS2 and PKIS1 (Table S16). The physicochemical basis had mean out-of-fold target $R^2 = 0.107$ and reduced the operational mean-r^2 contrast by 0.234. A seven-dimensional unnormalised count-Morgan random projection gave 0.090 and 0.238 for the locked seed. Across 20 count-Morgan seeds, operational reduction ranged from 0.176 to 0.260; at least 17 of 20 component maps equalled or exceeded the physicochemical map's experimental agreement in every panel. The fixed basis therefore does not distinguish a privileged physicochemical explanation from a more generic target-indexed ligand-feature basis.

The negative controls show why the scale-free map statistic cannot repair that identifiability problem. A stable seven-dimensional ligand-identity hash had mean out-of-fold target R^2 effectively zero and changed the operational contrast by only 0.00002, yet its fitted-component map correlated 0.103–0.348 with the four experimental maps. Twenty row permutations of the same standardised physicochemical feature cloud also had effectively zero held-out predictive performance, while their scale-free component-map agreements spanned -0.072 to 0.564 depending on panel and seed (Table S17). Each basis is fitted separately to every target, and the subsequent target–target correlation discards component amplitude. A random or nonpredictive sketch can therefore inherit target covariance from its fitted coefficient matrix. This is a scale-free random-sketch artefact of target-specific fitting followed by correlation. Component-map concordance is not a mediation estimate, a share of experimental structure, or evidence that the named ligand features orient biological selectivity.

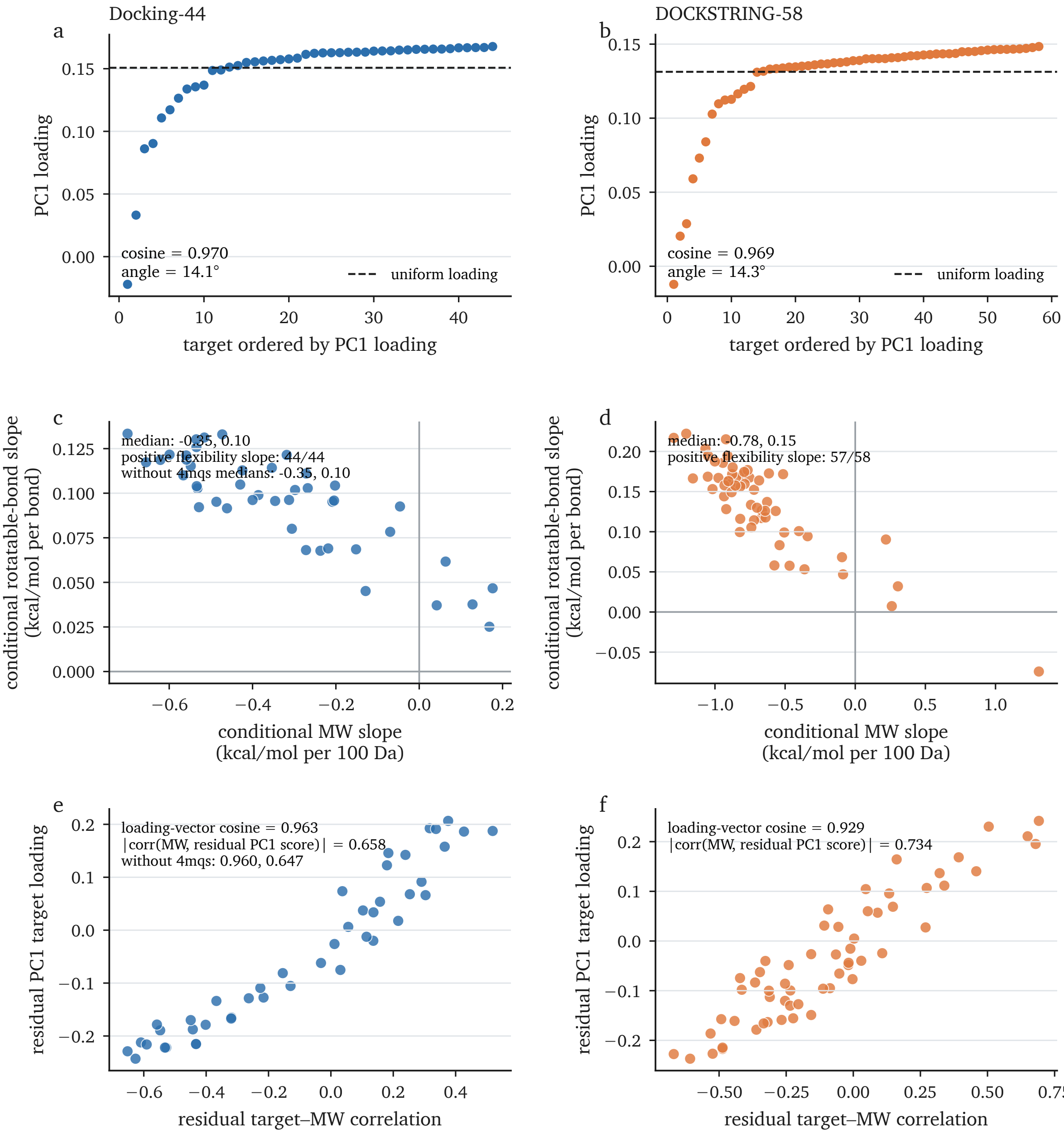

Figure S5: The shared axis is nearly uniform, whereas the leading residual mode tracks target-specific size dependence. **a,b**, Sorted raw PC1 target loadings and the uniform-loading reference. **c,d**, Conditional molecular-weight and RDKit rotatable-bond slopes for each target. **e,f**, Alignment of target-wise residual molecular-weight correlations with residual PC1 target loadings. Component signs are oriented by the molecular-weight vector and have no intrinsic meaning.

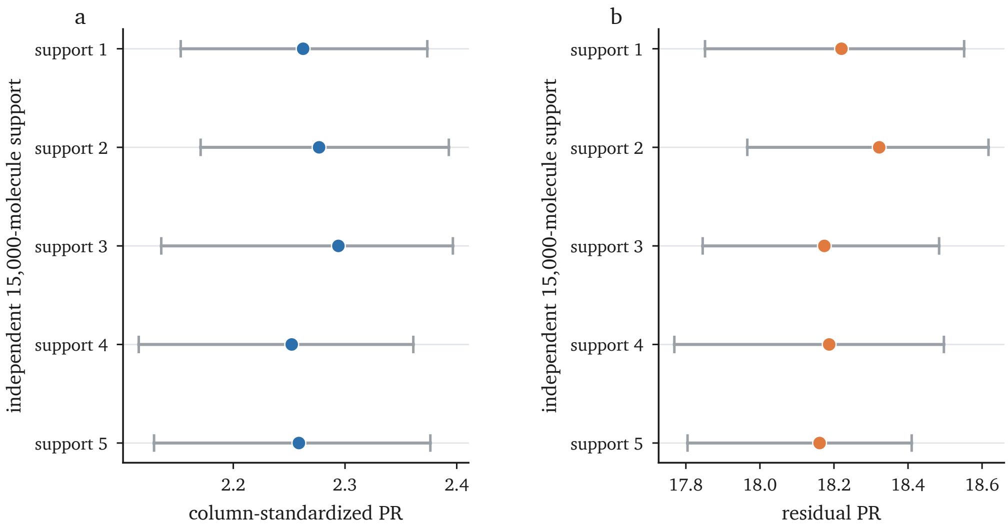

Figure S6: DOCKSTRING chemical-support sensitivity. Each row is an independently seeded, score-blind 15,000-molecule support. Points are support-specific PR dimensions; intervals are 95% Bemis–Murcko scaffold-cluster bootstrap intervals from 100 replicates. **a**, Column-standardized surface. **b**, Two-way-residual surface. All estimates are conditional on the same 58 targets.

S4. Experimental target-pair geometry

The complete factorial comparison prevents a joint change of predictor and endpoint from being attributed to one side. Against the fixed centred experimental geometry, changing raw to centred docking geometry increased Spearman concordance by only 0.001–0.045 across the four panels, with paired target-label QAP probabilities 0.126–0.462 and transformation-matched probabilities 0.073–0.690 (Table S7). Against the fixed uncentred experimental geometry, the same docking transformation gave increments of 0.168, 0.249, 0.218 and 0.264 for DAVIS, PKIS2, PKIS1 and KiRHub, with target-label probabilities 0.01512, 0.00152, 0.00536 and 0.00008 and transformation-matched probabilities 0.00300, 0.00050, 0.00050 and 0.00050. Thus the apparent docking-side gain is conditional on the experimental endpoint. The matched null begins at near-zero docking–experiment concordance rather than the already positive raw agreement with centred experiment, so its gain comparison is unconditional; Table S7 therefore prints both the full-reference and matched-support increments. Sequential percentages depend on transformation order; path-averaged experimental shares were 55.8–70.9% and are reported only descriptively (Table S8). These shares also depend on the Spearman-correlation scale; a nonlinear transform such as Fisher's z changes the allocation without changing the four underlying comparisons. The negative interaction terms show why no unique attribution of the joint raw/raw-to-centred/centred change is available.

The four experimental centred geometries showed strong cross-panel agreement on the shared target support: mean pairwise Spearman concordance was 0.763, compared with 0.632 for the uncentred geometries (Table S10). A global target-label network QAP supported the centred-minus-raw cross-panel increment ($p = 0.0020$) because it asks whether corresponding target labels align across panels. A transformation-matched independent-column null instead gave $p = 0.947$ because it asks whether the observed increment exceeds the correlation change mechanically induced by centring. The two probabilities therefore answer different questions. In particular, the matched null starts at mean raw agreement -0.0002, rather than the observed 0.632, and has mean centred agreement 0.179; its increment is therefore not conditioned on the already strong observed raw agreement. The absolute centred agreement nevertheless exceeded the same matched null ($p = 0.00050$). Thus the centred geometries agree, whereas whether their increment is exceptional depends on the null hypothesis.

DAVIS has an important measurement boundary. Of its 1,440 fixed-panel cells, 965 (67.0%)

are at the released 10,000-nM Kd upper bound, which becomes a pKd floor of 5; target-wise fractions range from 33.3–97.2%, and 4 targets exceed 90%. The continuous and binary above-floor-status target geometries nevertheless correlated at $\rho = 0.885$. Centred DOCKSTRING agreed with the binary geometry at $\rho = 0.292$ ($p = 0.00426$), and the association remained positive when targets were required to have at least 5, 10 or 15 values above the floor (Table S11). Replacing DAVIS continuous correlations by the binary-status geometry retained 13 of the 15 fixed pairs, defined 16 positives, and was recovered by centred KiRHub geometry with AUROC 0.956 and AP 0.673. This supports a coarse co-response or measurement-limit-status geometry, not a finely resolved quantitative DAVIS affinity geometry.

The earlier 74-ligand by 6-target ChEMBL comparison remains a matched-support spectral control (Table S25). Before two-way centring, experimental PR exceeded docking PR by 2.006 (paired ligand-bootstrap 95% interval 1.174–2.673). After centring, the difference was 0.483 and its interval included zero (-0.071–0.921). This small panel does not support a general claim about the dimension of experimental affinity surfaces.

S5. Structural interpretation boundary

Sequence identity, KLIFS pocket identity, raw Vina geometry and centred Vina geometry were evaluated against the same locked 15-of-190 pair endpoint. Sequence and pocket similarity gave AUROC 0.718 and 0.708, compared with 0.597 and 0.710 for raw and centred Vina (Table S13). These point estimates make protein similarity an essential baseline; they do not identify a docking-specific mechanism.

Because the endpoint was fixed from DAVIS, PKIS2 and PKIS1 before KiRHub was inspected, KiRHub supplied a temporally ordered external-panel check, although it was not preregistered. Raw and centred KiRHub experimental geometries retrieved the 15 pairs with AUROC 0.803 and 0.963 and average precision 0.597 and 0.694, respectively. Each centred absolute metric had target-label QAP $p = 2 \times 10^{-5}$. Paired target-label relabelling supported the centred-minus-raw gains (AUROC $\Delta = 0.160$, $p = 0.04928$; AP $\Delta = 0.097$, $p = 0.02306$; Holm-adjusted $p = 0.04928$ and 0.04612). Under the independent-column transformation-matched null, those gains did not exceed the mechanical centring gain ($p = 0.748$ and 0.997), although each absolute centred metric exceeded that null ($p = 0.00050$; Table S13). The matched raw AUROC and AP baselines were near chance rather than the much higher observed raw metrics, so these gain probabilities do not condition on the already aligned raw endpoint.

Continuous partial-rank models adjusted centred Vina geometry for receptor-domain sequence identity and KLIFS pocket identity. Unrestricted target-label QAP gave partial Spearman 0.225 for the three-panel mean and 0.280 for KiRHub. When labels were permuted only within KLIFS groups, the corresponding probabilities were 0.197 and 0.166 (Table S20). Coarse kinase organization is compatible with the observed map, while an additional within-group Vina component was not resolved.

As an orthogonal, heavily censored check, the Secondary Pharmacology Database release supplied 59 of 66 possible pairs among 12 mapped target groups for the primary right-censored endpoint; 91.6% of cells were censored. Centred Vina improved the released- bound point estimate from 0.149 to 0.333 ($p = 0.046$), and the 30-μM sensitivity from 0.180 to 0.409 ($p = 0.020$), but not the 10-μM analysis (0.348 to 0.395; $p = 0.347$). Family-preserving tests were unresolved. Specifically, family-preserving QAP for the centred-minus-raw gain gave one-sided $p = 0.729$, 0.939 and 0.714 for the released-bound, 10-μM and 30-μM endpoints, respectively. This result is a censoring sensitivity, not an independent quantitative-affinity validation.

S6. Chemical and target support

For DAVIS, PKIS2 and PKIS1, all experimental Standard-InChI connectivity blocks were excluded from the 20-target DOCKSTRING reference. The union contained 1,050 blocks and removed 481 rows, leaving 259,579 reference ligands. Excluding an additional 3,039 rows that shared one of 575 cyclic Murcko scaffolds with these panels changed centred geometry concordance by at most 0.0017. KiRHub compound structures were unavailable, so this exclusion could not be performed for KiRHub. Five score-blind 15,000-row reference supports reproduced each full-reference centred concordance within narrow ranges (Table S9).

Chemical context nevertheless matters. Target maps from the lower and upper molecular-weight quartiles had Spearman concordance 0.333 for PKIS2 and 0.547 for PKIS1, below matched random disjoint splits ($p = 0.001$ for each); the smaller DAVIS split was unresolved (Table S21). Changing the row-mean centring panel from the local 20 targets to all 58 DOCKSTRING targets changed experimental concordance by -0.008 to -0.019, with paired QAP probabilities 0.678–0.925. The two resulting docking maps themselves correlated 0.973.

The same post-hoc split was more severe on the full Docking-44 support and a fixed, score-blind 15,000-ligand DOCKSTRING support. The low/high-MW residual maps correlated 0.205 and 0.351, with paired chemical-group bootstrap intervals 0.185–0.219 and 0.326–0.363 (Table S18). By contrast, chemically disjoint supports constructed to have closely matched MW distributions correlated 0.991 and 0.994. The observed splits changed the signs of 42.5% and 36.7% of target-pair edges. The cross-band effect did not depend on row centring: raw-map concordances were 0.227 (cluster-bootstrap interval 0.209–0.246) and 0.409 (0.384–0.432), versus raw MW-matched-control means of 0.995 and 0.998.

We also calibrated the consequence of imposing either MW restriction without crossing the domain boundary. MW-stratified whole-chemical-group-disjoint halves within the low and high bands gave residual-map means of 0.980 (repeated-split range 0.974– 0.984) and 0.978 (0.967– 0.985) for Docking-44. DOCKSTRING-58 gave 0.978 (0.974– 0.982) and 0.966 (0.957–0.972). Corresponding raw means were 0.989/0.989 and 0.991/0.979. These central 95% repeated-split ranges quantify composition sensitivity rather than population uncertainty. Both restricted-domain maps were internally reproducible; the low/high contrast is not a generic consequence of estimating correlations after truncating an MW distribution.

Size-matched random disjoint supports correlated 0.991 and 0.990, excluding quartile sample size as the main explanation. Strict group-held-out descriptor removal raised the low/high-MW agreement to 0.489 and 0.437 but did not restore matched-domain agreement. The DOCKSTRING result was stable across six independently seeded 15,000-ligand supports (0.325–0.361). Molecular weight was selected after preliminary inspection, so these analyses define a transport boundary rather than a confirmatory descriptor-specific effect.

As a post-hoc family sensitivity, we repeated the inclusive 25th/75th-percentile-threshold comparison for every descriptor in the already inspected seven-descriptor set (Table S19); all observations tied at either boundary were retained. The three lowest residual-map agreements in both matrices were the correlated molecular-size descriptors: Labute ASA, molecular weight and heavy-atom count. The remaining TPSA, cLogP, rotatable-bond and ring-count splits also produced agreements below the same dataset-level MW-matched chemical-group-disjoint reference, not descriptor-matched nulls; thus the ordering does not isolate molecular weight or any other unique mechanism. For the molecular-weight split, residual PR changed from 13.625 to 10.147 between the low- and high-MW Docking-44 domains, but from 19.552 to 23.800 in DOCKSTRING-58. This opposite direction precludes a common monotone interpretation of effective dimension as a function of molecular size or as a docking-quality signal.

Sampling recovery asks a narrower question: how many ligands are needed to reconstruct the target map of the same source library? At 200 ligands, mean recovery Spearman was 0.942 for Docking-44 and 0.918 for DOCKSTRING-58; at 500 it was 0.976 and 0.965 (Table S22). Chemical-group-held-out recovery at 200 ligands remained 0.927 for Docking-44 and 0.916 for DOCKSTRING-58. These results support efficient estimation of an existing source-library map, not transfer to a new chemical distribution or biological validation. The 200-ligand 2.5–97.5% replicate ranges were 0.064 wide for Docking-44 and 0.053 for DOCKSTRING-58. Because that sample-to-sample spread is comparable to or larger than the 0.001–0.045 docking-side increments against the fixed centred endpoints, this sampling result does not support fine discrimination of small map or method differences.

Murcko clustering used canonical non-empty cyclic Bemis–Murcko scaffolds; acyclic compounds were grouped by connectivity identity. Butina clustering used RDKit Morgan radius-2, 2,048-bit fingerprints with Tanimoto similarity at least 0.65 (distance at most 0.35). Singletons remained one-member clusters. Cluster bootstraps sampled clusters with replacement and included every ligand in each selected cluster; clusters were not reduced to representatives. The broad primary benchmark contained 2,815 Murcko and 2,312 Butina clusters; the legacy 137-ligand Docking-44 sensitivity contained 95 and 121, respectively.

S7. Ligand-wise observed-pair ranking

Subtracting the same ligand row mean from every target leaves all within-ligand target differences unchanged:

$$(x_{ij} - \bar{x}_{i\cdot}) - (x_{ik} - \bar{x}_{i\cdot}) = x_{ij} - x_{ik}.$$

Target centring can change rankings through target-specific offsets, and subsequent target-specific residual scaling can change them further. In the broad primary benchmark, target-centred and two-way-centred unscaled scores both gave mean observed-pair concordance 0.533 (Table S26); the equality is algebraic, not an empirical null result.

The broad DOCKSTRING–ChEMBL benchmark retained 6,557 ligands over 31 targets and 18,020 observed cells; 6,480 ligands contributed 25,214 non-tied comparisons. Absolute Vina, column-standardised Vina and scaled two-way residual scores gave mean per-ligand concordances 0.537, 0.530 and 0.529. Residual minus absolute was -0.008, with a conservative chemical-cluster 95% interval of -0.022 to 0.006 and a delete-one-target jackknife interval of -0.056 to

0.030 (Tables S26 and S27). The scaffold-cross-fitted cohort outcome prior scored 0.604, above every docking representation, but is fitted to the benchmark outcomes and is not a deployable predictor. Absolute Vina was the highest of the docking-derived representations; the data did not resolve a scaled-residual advantage.

Restricting the endpoint exposed heterogeneity rather than a common transformation effect (Table S27). The human binding Ki/Kd arm retained 1,056 ligands and gave absolute and residual concordances 0.559 and 0.520; residual minus absolute was -0.040 (chemical-cluster 95% interval -0.091 to 0.003; target-jackknife interval -0.133 to 0.029). Each cell in this arm remains the median over all eligible Ki and Kd records. Of its 3,955 informative comparisons, 3,610 (91.3%) linked two unambiguous cells of the same endpoint type, 33 (0.8%) linked an unambiguous Ki cell to an unambiguous Kd cell, and 312 (7.9%) involved at least one cell pooled across both record types. Thus 99.1% of the 3,643 endpoint-unambiguous comparisons were same-endpoint; dual-type cells were not forced into either category. The single-Ki arm gave -0.033 (-0.069 to -0.005). Absolute Vina scored 0.488 in the smaller Kd arm, whose residual-minus-absolute contrast was $+0.038$ (-0.008 to 0.080). IC50 and EC50 contrasts were -0.006 and $+0.013$, with both intervals spanning zero. The Ki interval excludes zero only for chemical resampling conditional on its fixed 31-target panel; no endpoint-specific target jackknife was run for the four single-endpoint arms, which are exploratory and unadjusted for multiplicity.

Dense exact-structure checks led to the same aggregate boundary (Table S30). DAVIS retained 56 of 59 matched ligands and 6,065 informative comparisons; PKIS2 retained all 154 ligands and 16,417 comparisons after its 10-percentage-point margin. Residual-minus-absolute differences were -0.009 and -0.003; both chemical-cluster and delete-one-target intervals included zero. Absolute-minus-target-prior increments were also unresolved. Path decomposition showed why the net contrast is small: residual-SD scaling reduced concordance, while adding the ligand-row term under unequal target scales increased it. The individual path intervals are correlated and unadjusted for multiplicity, so they localise the cancellation rather than establish independent effects.

Outcome-conditioned dense strata were not uniform (Table S31). DAVIS both-uncensored and PKIS2 both-active comparisons favoured the residual rule by 0.044 and 0.052, whereas DAVIS floor-versus-uncensored comparisons gave -0.013. These strata use the outcomes to define membership and are exploratory. In the DAVIS both-uncensored stratum, absolute Vina was below chance (0.459) and the residual rule was near chance (0.503), so the positive difference is not evidence of useful discrimination.

The equivalence-margin curves are retained only as post-hoc Supplementary sensitivity analyses (Table S28). No margin was preregistered. A margin of 0.03 contains the conservative 90% chemical-cluster intervals for the three aggregate benchmarks, but the outcome-conditioned DAVIS both-uncensored and PKIS2 both-active strata do not pass any margin on the 0.010–0.060 grid. These curves neither establish global equivalence nor alter the primary "no resolved advantage" interpretation.

Finally, the 137-ligand by 38-target Docking-44–ChEMBL analysis is a legacy sparse sensitivity (Table S29). It gave 0.566, 0.546 and 0.564 for absolute, column-standardised and scaled residual Vina on 2,522 comparisons. Its median co-observation support was three ligands per target pair, and 44 ligands had exactly two measured targets. Changing only the ligand-row mean from all 44 Docking-44 targets to the local 38-target subset altered 31 predictions in 19 ligands (Table S32); the associated coverage and assay sensitivities are in Table S33. This small support is not used as the primary operational estimate.

S8. Reusable audit and reproducibility

The reusable output of this study is the decision-oriented audit stated in the main text: separate redundancy, matched nulls, panel composition, experimental target-pair geometry and ligand-wise ranking, and report each with its own boundary. The command-line tool computes the spectral portion for a generic ligand-by-target matrix. Dataset-specific scripts reproduce the external geometry, structural controls, support recovery and ranking analyses from their frozen or checksum-verified inputs.

All numerical tables below are generated by `analysis/make_supplement_tables.py`. Headline values in both PDFs are generated from `results/manuscript_evidence.json` through `analysis/make_reported_results.py`; the source hashes and analysis contracts are stored in the same ledger. Randomized analyses use fixed recorded seeds. Frozen redistributable inputs are covered by `data_manifest.csv`; non-redistributable source files have checksum-verified retrieval instructions and contribute only derived, non-row-level artifacts to the release.

Table S1: Large-matrix source support, analysis support and spectral summaries. Source missingness is reported before the stated handling; the analyzed matrices are complete after Docking-44 imputation or DOCKSTRING row exclusion.

Matrix	source N	analysis N	P	source missing (%)	raw PR	residual PR	raw PC1	residual PC1	raw $\overline{\lvert r \rvert}$	residual $\overline{\lvert r \rvert}$
Docking-44	12,651	12,651	44	1.466	1.834	9.303	73.3%	27.4%	0.694	0.246
DOCKSTRING-58	260,155	260,060	58	0.002	2.266	18.206	65.9%	17.8%	0.622	0.155

Table S2: Large-matrix residual-spectrum sensitivity to missing-score and score-clipping handling.

Method	support	N	P	residual PR	interpretation
target mean imputation	all targets	12,651	44	9.303	missing scores equal the observed target mean
target median imputation	all targets	12,651	44	9.528	missing scores equal the observed target median
observed-cell additive least-squares imputation	all targets	12,651	44	9.663	Each missing ligand-target interaction residual is set to zero after fitting the additive ligand and target effects on observed cells.
complete-case deletion	all targets	9,297	44	13.970	changes chemical support and is not a like-for-like imputation analysis
primary mean-imputed matrix restricted to complete rows	all targets	9,297	44	13.970	equals complete-case input exactly because these rows contain no missing cells
target mean imputation after target removal	without 4mqs	12,651	43	9.433	removes the target with the largest missing-cell count
target median imputation after target removal	without 4mqs	12,651	43	9.594	removes the target with the largest missing-cell count
complete cases after target removal	without 4mqs	11,593	43	11.445	changes both chemical and target support
retain positive DOCKSTRING scores	complete rows	260,060	58	18.362	retain 5,393 positive cells instead of clipping them at zero

Table S3: Correlation- and covariance-spectrum sensitivity.

Matrix	estimand	raw PR	residual PR	raw PC1 (%)	residual PC1 (%)	raw PCs_{90}	residual PCs_{90}
Docking-44	correlation	1.834	9.303	73.3	27.4	8	29
Docking-44	covariance	1.915	6.911	71.6	31.3	8	21
DOCKSTRING-58	correlation	2.266	18.206	65.9	17.8	22	43
DOCKSTRING-58	covariance	2.474	8.235	62.4	31.9	20	37

Table S4: Residual null models on fixed 12,000-ligand supports. Within each matrix, all three nulls use the identical deterministic ligand support; its SHA-256 row-index fingerprint is stored in the machine-readable evidence ledger.

Matrix	null	observed residual PR	null median PR	draws	lower-tail p
Docking-44	additive Gaussian	9.295	42.400	500	0.002
Docking-44	empirical column permutation	9.295	42.403	500	0.002
Docking-44	row-norm preserving	9.295	42.628	500	0.002
DOCKSTRING-58	additive Gaussian	18.158	56.482	500	0.002
DOCKSTRING-58	empirical column permutation	18.158	56.481	500	0.002
DOCKSTRING-58	row-norm preserving	18.158	56.612	500	0.002

Table S5: Leave-one-target-out spectral composition sensitivity.

Matrix	surface	full PR	minimum	median	maximum	most influential deletion
Docking-44	raw	1.834	1.756	1.846	1.853	4mqs
Docking-44	residual	9.303	8.956	9.206	9.872	8e9y
DOCKSTRING-58	raw	2.266	2.218	2.297	2.311	AR
DOCKSTRING-58	residual	18.206	17.662	17.931	20.045	MAOB

Table S6: Random and family-stratified target-subsampling sensitivity.

Matrix	design	targets	raw median	raw 2.5%	raw 97.5%	residual median	residual 2.5–97.5%
Docking-44	random	4	1.559	1.166	2.569	2.537	1.825–2.961
Docking-44	random	6	1.629	1.237	2.517	3.680	2.453–4.583
Docking-44	random	8	1.649	1.281	2.283	4.615	3.184–5.670
Docking-44	random	12	1.701	1.340	2.336	5.938	4.283–7.602
Docking-44	random	20	1.789	1.496	2.146	7.338	5.693–9.389
Docking-44	random	32	1.841	1.649	1.993	8.547	7.447–9.994
Docking-44	random	44	1.834	1.834	1.834	9.303	9.303–9.303
Docking-44	family-stratified	12	1.876	1.609	2.334	5.834	4.410–7.300
Docking-44	family-stratified	20	1.852	1.625	2.169	7.375	5.932–9.045
Docking-44	family-stratified	32	1.835	1.663	1.978	8.517	7.269–10.014
Docking-44	family-stratified	44	1.834	1.834	1.834	9.303	9.303–9.303
DOCKSTRING-58	random	4	1.645	1.369	2.721	2.844	1.873–2.988
DOCKSTRING-58	random	6	1.945	1.490	2.805	4.342	2.765–4.845
DOCKSTRING-58	random	8	2.004	1.551	2.849	5.737	3.262–6.505
DOCKSTRING-58	random	12	2.074	1.664	2.732	8.051	5.198–9.442
DOCKSTRING-58	random	20	2.180	1.746	2.790	11.196	7.669–14.391
DOCKSTRING-58	random	32	2.233	1.958	2.537	14.367	10.905–18.012
DOCKSTRING-58	random	44	2.240	2.043	2.400	16.467	13.717–20.031
DOCKSTRING-58	random	58	2.266	2.266	2.266	18.206	18.206–18.206
DOCKSTRING-58	family-stratified	6	1.947	1.548	2.545	4.538	3.021–4.847
DOCKSTRING-58	family-stratified	8	2.024	1.617	2.674	5.983	3.718–6.594
DOCKSTRING-58	family-stratified	12	2.093	1.657	2.628	8.363	5.384–9.713
DOCKSTRING-58	family-stratified	20	2.217	1.875	2.624	11.401	7.414–14.186
DOCKSTRING-58	family-stratified	32	2.247	2.051	2.548	14.612	11.227–17.975
DOCKSTRING-58	family-stratified	44	2.273	2.087	2.429	16.850	14.007–19.743
DOCKSTRING-58	family-stratified	58	2.266	2.266	2.266	18.206	18.206–18.206

Table S7: Complete docking-by-experimental estimand decomposition on the common 20 kinases. Each $\Delta_D^{F/M}$ cell gives the full-reference increment followed by the increment on the fixed 15,000-ligand support; p_L denotes the paired target-label QAP for the first and p_M the transformation-matched independent-column null for the second. Differences were calculated from unrounded correlations and can therefore differ slightly from subtraction of the displayed entries.

Panel	N	D_R/E_R	D_R/E_C	D_C/E_R	D_C/E_C	$\Delta_D^{F/M} \mid E_R$	p_L	p_M	$\Delta_D^{F/M} \mid E_C$	p_L	p_M	interaction
DAVIS	72	0.050	0.263	0.219	0.280	+0.168/+0.162	0.01512	0.00300	+0.016/+0.010	0.335	0.690	-0.152
PKIS2	645	-0.215	0.243	0.033	0.287	+0.249/+0.243	0.00152	0.00050	+0.044/+0.034	0.146	0.172	-0.205
PKIS1	360	-0.159	0.173	0.059	0.174	+0.218/+0.215	0.00536	0.00050	+0.001/-0.006	0.462	0.513	-0.217
KiRHub	92	-0.029	0.276	0.235	0.322	+0.264/+0.261	0.00008	0.00050	+0.045/+0.041	0.126	0.073	-0.219

Table S8: Path-dependent allocation of the joint raw/raw-to-centred/centred change.

Panel	joint change	experiment-first Δ_E	final Δ_D	sequential E fraction	path-averaged E share	path-averaged D share
DAVIS	0.230	0.213	0.016	92.9%	0.137 (59.8%)	0.092 (40.2%)
PKIS2	0.502	0.459	0.044	91.3%	0.356 (70.9%)	0.146 (29.1%)
PKIS1	0.333	0.332	0.001	99.7%	0.223 (67.1%)	0.109 (32.9%)
KiRHub	0.351	0.305	0.045	87.0%	0.196 (55.8%)	0.155 (44.2%)

Table S9: Strict common-20 reference-library sensitivity for centred docking versus centred experiment.

Panel	full 259,579-row reference	same-scaffold-excluded	change	five 15,000-row supports
DAVIS	0.280	0.279	-0.0009	0.271–0.285
PKIS2	0.287	0.287	-0.0006	0.277–0.290
PKIS1	0.174	0.172	-0.0017	0.167–0.176
KiRHub	0.322	–	–	0.312–0.326

Table S10: Four-panel experimental geometry agreement under complementary null models.

Quantity	raw	centred	interpretation
Mean of six pairwise panel concordances	0.632	0.763	absolute centred target-label QAP $p = 2.0e-05$; transformation-matched absolute $p = 5.0e-04$
Centred minus raw	+0.131		global target-label network QAP $p = 0.002$; transformation-matched independent-column null $p = 0.947$
Transformation-matched null mean	-0.000	0.179	mean mechanical increment +0.179; its raw baseline is not conditioned on the observed 0.632
KiRHub versus mean of DAVIS/PKIS2/PKIS1	–	0.834	0.817 after removing 11 exact-name overlaps; 0.802 after sequence adjustment

Table S11: Censoring-aware DAVIS sensitivity on the fixed kinase panel. Each row compares centred DOCKSTRING target-pair geometry with the stated DAVIS geometry; probabilities use complete target-label permutations. The binary endpoint records whether Kd was measured below the released 10,000-nM limit, not quantitative affinity.

DAVIS representation	targets	Spearman ρ	QAP p
continuous pKd, all targets	20	0.280	0.005
binary above-pKd-5-floor status, all targets	20	0.292	0.004
continuous pKd; at least 5 values above floor per target	18	0.255	0.010
continuous pKd; at least 10 values above floor per target	16	0.210	0.030
continuous pKd; at least 15 values above floor per target	12	0.229	0.049

Table S12: Fixed 15-of-190 experimental target-pair endpoint, defined from DAVIS, PKIS2 and PKIS1 before KiRHub inspection; the ordering was not preregistered.

Target A	Target B	panels in top decile	DAVIS percentile	PKIS2 percentile	PKIS1 percentile	mean percentile
ABL1	KDR	2	0.434	0.955	0.934	0.775
ABL1	LCK	3	0.982	0.934	0.971	0.962
ABL1	SRC	2	0.961	0.887	0.982	0.943
AKT1	AKT2	3	0.997	0.997	0.997	0.997
AKT1	MAPK1	2	0.934	0.950	0.850	0.911
AKT1	MAPKAPK2	2	0.955	0.876	0.961	0.931
AKT1	ROCK1	3	0.945	0.961	0.908	0.938
AKT2	MAPK1	2	0.939	0.945	0.861	0.915
AKT2	MAPKAPK2	3	0.966	0.966	0.955	0.962
CSF1R	KDR	2	0.929	0.982	0.792	0.901
CSF1R	KIT	3	0.992	0.992	0.976	0.987
IGF1R	MAPK1	2	0.918	0.939	0.803	0.887
KDR	KIT	2	0.971	0.971	0.882	0.941
LCK	SRC	3	0.976	0.987	0.987	0.983
MAPK1	MAPKAPK2	3	0.987	0.976	0.966	0.976

Table S13: KiRHub evaluation and structural controls for the fixed 15-of-190 target-pair endpoint. The endpoint was defined before KiRHub inspection but not preregistered. Absolute-predictor rows use target-label QAP. The contrast rows show how the same centred-minus-raw increment is assessed under target-label and transformation-matched nulls; the absolute centred KiRHub predictor also exceeded the transformation-matched null ($p = 0.00050$ for both metrics).

Quantity	AUROC/value	AUROC null p	AP/value	AP null p
raw KiRHub experimental geometry	0.803	0.00014	0.597	0.00002
centred KiRHub experimental geometry	0.963	0.00002	0.694	0.00002
centred minus raw KiRHub, target-label QAP	+0.160	0.04928	+0.097	0.02306
centred minus raw KiRHub, transformation-matched null	+0.160	0.74763	+0.097	0.99650
Holm-adjusted target-label gain probabilities	–	0.04928	–	0.04612
receptor-domain sequence identity	0.718	0.007	0.426	2.0e-05
KLIFS pocket identity	0.708	0.010	0.486	2.0e-05
same KLIFS group	0.651	0.043	0.126	0.016
same KLIFS family	0.597	0.001	0.192	0.002
raw Vina geometry	0.597	0.155	0.236	0.011
centred Vina geometry	0.710	0.007	0.301	0.001

Table S14: Chemical-group-held-out predictive decomposition of the residual docking surfaces. Every target offset, residual target scale, descriptor scale and regression coefficient was fitted within the training fold. The seven descriptors were heavy-atom count, molecular weight, Labute ASA, TPSA, cLogP, rotatable-bond count and ring count. The change in mean squared target correlation is predictive, not causal attribution.

Matrix	ligands	chemical groups	residual PR before	PR after descriptor removal	mean r^2 reduction	residual-PC1 out-of-fold R^2
Docking-44	12,651	8,150	9.294	14.746	46.9%	0.586
DOCKSTRING-58	15,000	11,905	17.923	28.466	53.6%	0.687

Table S15: Spectral concentration under six non-Vina scoring functions on identical fixed DOCKSTRING poses. Every scorer sees the same 2,000-ligand by 9-target pose block; only the scoring function changes. The four prediction criteria were fixed before the non-Vina scores were inspected: raw PC1 fraction at least 0.50, cosine to the uniform target direction at least 0.95 with no negative loading, raw participation ratio no greater than half the target count, and a centring-induced participation-ratio increase whose 95% cluster-bootstrap interval excludes zero. These criteria are a weak bar: a one-factor surface with no target identity satisfies all four, so meeting them evidences a shared positive ligand axis and nothing finer. Two margins sit inside bootstrap noise: released Vina passes the cosine criterion at 0.9501, and PLECscore exceeds the participation-ratio threshold at the point estimate with an interval that spans it, whereas its PC1 failure holds in every replicate. Scores are not clipped, because clipping only the Vina-family columns would be asymmetric. On a separate 512-ligand fixed-pose within-Vina attribution, gauss2 supplied 0.715 of the shared-axis covariance and 0.525 of the residual Frobenius inner product. These are signed arithmetic attributions without resampling uncertainty; term deletion on Vina-selected poses is not an independent redocking ablation.

Scorer	family	raw PC1	raw PR	centred PR	PR increase	uniform cosine	criteria met
released vina	Vina family	57.3%	2.74	5.12	2.38	0.9501	all four
smina vina	Vina family	57.7%	2.71	5.13	2.42	0.9546	all four
vinardo	empirical	56.7%	2.81	5.74	2.93	0.9668	all four
rfscore v1	machine learned	63.5%	2.34	6.44	4.10	0.9587	all four
rfscore v2	machine learned	66.9%	2.13	5.65	3.52	0.9535	all four
rfscore v3	machine learned	63.0%	2.37	6.00	3.63	0.9562	all four
nnscore	machine learned	52.6%	3.21	6.79	3.59	0.9730	all four
plecscore linear	machine learned	41.4%	4.59	7.47	2.87	0.9858	PC1 fails

Table S16: Rank-matched ligand-feature controls on the fixed common-20 kinase support. Each seven-dimensional basis receives separate target-specific coefficients in five chemical-group-held-out folds. Mean out-of-fold target R^2 measures held-out predictive performance. Operational reduction is $1 - \overline{r^2}_{\mathrm{error}}/\overline{r^2}_{\mathrm{observed}}$; it is non-additive and is not a share of correlation, covariance or variance explained. The four final columns are scale-free Spearman agreements between the fitted-component target-correlation map and each experimental map over the same 190 edges. The hash arm has negligible held-out predictive performance but appreciable map agreement, so component-map agreement alone cannot identify a feature-specific mechanism or mediation path.

Seven-dimensional basis	mean OOF target R^2	operational reduction	DAVIS ρ	PKIS2 ρ	PKIS1 ρ	KiRHub ρ
physicochemical descriptors	0.1068	23.361%	0.256	0.218	0.097	0.288
count-Morgan random projection (locked seed)	0.0903	23.765%	0.299	0.279	0.139	0.359
stable ligand-identity hash nuisance	-3.63×10^{-5}	0.002%	0.248	0.250	0.103	0.348

Table S17: Algorithmic basis sensitivity for the rank-matched controls. The count-Morgan column spans 20 deterministic unnormalised count-fingerprint projections; the row-permuted column spans 20 deterministic permutations of the same standardised physicochemical feature cloud, which destroy the ligand–feature assignment. Ranges are across fixed seeds, not confidence intervals or chemical-space uncertainty. A scale-free component map can remain concordant when held-out predictive performance is effectively null because every nuisance basis is projected separately onto each target and correlation discards component scale. These rows therefore cannot be read as mediation or feature-specific biological attribution.

Quantity	physicochemical point	count-Morgan 20-seed range	row-permuted-physicochemical 20-seed range
mean OOF target R^2	0.107	0.081 – 0.098	-0.00005 – -0.00002
operational reduction	0.234	0.176 – 0.260	-0.00007 – 0.00007
component-map agreement: DAVIS	0.256	0.236 – 0.336	0.034 – 0.489
component-map agreement: PKIS2	0.218	0.238 – 0.322	0.025 – 0.452
component-map agreement: PKIS1	0.097	0.100 – 0.173	-0.072 – 0.401
component-map agreement: KiRHub	0.288	0.299 – 0.404	0.030 – 0.564

Table S18: Exploratory low- versus high-molecular-weight transfer of the docking target-pair maps. Parenthesized intervals for the cross-band and descriptor-adjusted contrasts are paired chemical-group bootstrap central 95% intervals. Within-band controls split the low- and high-MW restrictions separately, keep whole chemical groups disjoint, and report raw/residual means over 200 MW-stratified repeated splits; their full 2.5–97.5% composition-sensitivity ranges are in the accompanying machine-readable CSV and are not population confidence intervals. The size-matched control uses random disjoint supports matching the observed low/high row counts; the MW-matched control uses chemically disjoint supports with closely matched molecular-weight distributions. Descriptor adjustment is five-fold group-held-out and fold-local. The final column gives the residual cross-band range over six independently seeded fixed DOCKSTRING supports and is an algorithmic support sensitivity, not an interval.

Matrix	raw cross-band ρ (95%)	residual cross-band ρ (95%)	within-low raw/residual	within-high raw/residual	size-matched residual	MW-matched residual	adjusted residual ρ (95%)	six-seed residual range
Docking-44	0.227 (0.209–0.246)	0.205 (0.185–0.219)	0.989/0.980	0.989/0.978	0.991	0.991	0.489 (0.446–0.515)	–
DOCKSTRING-58	0.409 (0.384–0.432)	0.351 (0.326–0.363)	0.991/0.978	0.979/0.966	0.990	0.994	0.437 (0.406–0.458)	0.325–0.361

Table S19: Post-hoc descriptor-family sensitivity of residual-map transport. Each row compares groups at the inclusive 25th- and 75th-percentile thresholds of one of the seven already inspected descriptors on the row-centred residual surface; all boundary ties are retained. The common within-dataset MW-matched reference is the mean agreement between chemically disjoint supports with closely matched molecular-weight distributions; it is not separately descriptor-matched. All 14 observed agreements are below that dataset-level reference. The molecular-size family supplies the three lowest agreements in both panels, but this descriptive ordering neither identifies a unique descriptor mechanism nor constitutes multiplicity-adjusted inference.

Matrix	stratifier	family	low/high N	residual PR (low)	residual PR (high)	map ρ	sign flips (%)	MW-matched control ρ
Docking-44	heavy-atom count	molecular size	3,545/3,523	14.54	9.90	0.207	42.4	0.991
Docking-44	molecular weight	molecular size	3,163/3,163	13.62	10.15	0.205	42.5	0.991
Docking-44	Labute ASA	molecular size	3,163/3,163	14.16	10.16	0.200	44.0	0.991
Docking-44	TPSA	polarity	3,163/3,167	10.47	8.62	0.753	20.8	0.991
Docking-44	cLogP	lipophilicity	3,164/3,163	8.44	9.65	0.887	13.1	0.991
Docking-44	rotatable bonds	flexibility	3,736/3,168	8.74	9.31	0.789	20.3	0.991
Docking-44	ring count	ring topology	4,537/4,349	12.42	9.39	0.703	22.6	0.991
DOCKSTRING-58	heavy-atom count	molecular size	4,367/3,817	20.03	23.78	0.361	36.3	0.994
DOCKSTRING-58	molecular weight	molecular size	3,751/3,750	19.55	23.80	0.351	36.7	0.994
DOCKSTRING-58	Labute ASA	molecular size	3,750/3,750	20.06	23.62	0.322	37.7	0.994
DOCKSTRING-58	TPSA	polarity	3,755/3,750	17.12	19.61	0.787	19.7	0.994
DOCKSTRING-58	cLogP	lipophilicity	3,750/3,750	18.69	19.40	0.822	17.5	0.994
DOCKSTRING-58	rotatable bonds	flexibility	6,154/4,352	17.27	18.45	0.831	18.5	0.994
DOCKSTRING-58	ring count	ring topology	8,376/6,624	19.97	22.04	0.803	18.2	0.994

Table S20: Continuous Vina association after sequence and pocket adjustment.

Experimental endpoint	partial Spearman	unrestricted QAP p	within-KLIFS-group QAP p
mean of DAVIS/PKIS2/PKIS1	0.225	0.014	0.197
KiRHub	0.280	0.007	0.166

Table S21: Chemical-support and centering-panel sensitivities.

Panel	low/high-MW ρ	random mean	matched p	local-20 ρ	all-58 ρ	all-58 minus local
DAVIS	0.531	0.619	0.173	0.280	0.261	-0.019
PKIS2	0.333	0.845	1.0e-03	0.287	0.274	-0.013
PKIS1	0.547	0.833	1.0e-03	0.174	0.166	-0.008
KiRHub	–	–	–	0.322	0.302	-0.019

Table S22: Recovery of each full source-library target map from sampled ligands.

Matrix	ligands	mean Spearman	2.5–97.5% range	chemical-group-held-out mean
Docking-44	50	0.813	0.657–0.901	–
Docking-44	100	0.898	0.772–0.946	–
Docking-44	200	0.942	0.898–0.962	0.927
Docking-44	500	0.976	0.961–0.984	0.959
Docking-44	1,000	0.987	0.977–0.991	–
Docking-44	2,000	0.994	0.991–0.996	–
Docking-44	5,000	0.998	0.997–0.999	–
DOCKSTRING-58	50	0.766	0.665–0.822	–
DOCKSTRING-58	100	0.861	0.808–0.896	–
DOCKSTRING-58	200	0.918	0.885–0.938	0.916
DOCKSTRING-58	500	0.965	0.951–0.972	0.958
DOCKSTRING-58	1,000	0.982	0.977–0.986	–
DOCKSTRING-58	2,000	0.991	0.987–0.992	–
DOCKSTRING-58	5,000	0.996	0.995–0.997	–

Table S23: Overlap between the two independently sourced Vina panels.

Overlap definition	count	boundary
full Standard InChIKeys	582	exact structure layer
14-character connectivity blocks	804	merges stereochemical and protonation suffixes
non-empty Bemis–Murcko scaffolds	1,340	scaffold-level chemical overlap
mapped protein identities	9	7 map to human receptors
same receptor structures	3	exact structure and chain used in both panels

Table S24: Chemical overlap among the three experimental panels with released structures. Full keys are recomputed Standard InChIKeys; connectivity is the 14-character block; Murcko intersections exclude empty scaffolds.

Panel pair	N_A	N_B	full keys	connectivity	Murcko
DAVIS–PKIS2	72	645	3	6	10
DAVIS–PKIS1	72	360	1	1	5
PKIS2–PKIS1	645	360	7	7	43

Three-panel intersection: 0 full keys, 0 connectivity blocks and 3 non-empty Murcko scaffolds. KiRHub has 92 rows but no released structures; 11 exact normalized-name overlaps with DAVIS were removed in a sensitivity ($\rho = 0.817$). No PKIS–KiRHub identity claim is made.

Table S25: Matched 74-ligand by 6-target ChEMBL–docking spectral comparison.

Surface	docking PR	experiment PR	experiment minus docking	paired ligand-bootstrap 95% interval
column-standardized	1.827	3.833	+2.006	1.174 – 2.673
two-way residual	3.646	4.128	+0.483	-0.071 – 0.921

Table S26: Broad DOCKSTRING–ChEMBL observed-pair benchmark. The 6,557 retained ligands provide at least two observed targets; 6,480 have a non-tied pair and enter the mean per-ligand endpoint. The docking and external experimental priors are ligand-invariant target means fitted after excluding all evaluation ligands. The cohort outcome prior is refitted after excluding each evaluated ligand's complete Bemis–Murcko scaffold cluster and is a benchmark-fitted, non-deployable reference. Chemical intervals are the conservative union of Murcko- and Butina-cluster 95% intervals. Target-centred and two-way-centred unscaled scores are exactly rank-identical because they differ only by a ligand-wise constant. The final row is the paired absolute-Vina-minus-docking-prior contrast on the same evaluated ligands, not an additional representation.

Representation	mean per-ligand concordance	pair-weighted	top-1	evaluated ligands	pairs	chemical-cluster 95% interval
ligand-invariant docking target prior	0.524	0.514	0.444	6,480	25,214	0.495 – 0.552
ligand-invariant external experimental prior	0.480	0.495	0.417	6,480	25,214	0.451 – 0.511
scaffold-cross-fitted cohort outcome prior	0.604	0.572	0.521	6,480	25,214	0.573 – 0.632
absolute Vina	0.537	0.528	0.468	6,480	25,214	0.518 – 0.556
target-centred, unscaled	0.533	0.528	0.471	6,480	25,214	0.514 – 0.552
two-way-centred, unscaled	0.533	0.528	0.471	6,480	25,214	0.515 – 0.552
column-standardised Vina	0.530	0.526	0.467	6,480	25,214	0.511 – 0.550
target-centred, residual-SD scaled	0.526	0.519	0.463	6,480	25,214	0.507 – 0.545
scaled two-way residual	0.529	0.527	0.467	6,480	25,214	0.511 – 0.549
paired: absolute Vina minus docking target prior	+0.013	–	–	6,480	25,214	-0.011 – 0.038

Table S27: Endpoint-restricted DOCKSTRING–ChEMBL ranking arms. Every row retains the externally fitted docking transformation and changes only the experimental support/endpoint contract. The contrast is scaled two-way residual minus absolute Vina. Chemical 95% intervals are conservative unions over Murcko and Butina cluster bootstraps and remain conditional on target labels. Target intervals are delete-one-target jackknife normal approximations for the fixed panel; they address panel composition rather than chemical sampling. Separate Ki, Kd, IC50 and EC50 arms are exploratory and unadjusted for multiplicity. Panel c audits endpoint provenance without changing the human binding Ki/Kd estimand: each cell remains the pooled median over all eligible records, and any pair involving a cell with both Ki and Kd records is reported separately rather than assigned an artificial endpoint label.

a. Support and representation estimates

Endpoint contract	retained	evaluated	targets	cells	pairs	absolute	column-standardised	residual
all exact-relation endpoints (broad primary)	6,557	6,480	31	18,020	25,214	0.537	0.530	0.529
human binding Ki/Kd	1,056	1,049	31	2,783	3,955	0.559	0.512	0.520
single endpoint: Ki	1,608	1,594	31	5,891	12,895	0.549	0.512	0.516
single endpoint: Kd	160	160	24	763	2,534	0.488	0.532	0.526
single endpoint: IC50	4,701	4,647	30	11,301	9,963	0.539	0.536	0.533
single endpoint: EC50	439	424	17	882	435	0.581	0.620	0.594

b. Paired scaled-residual-minus-absolute contrasts

Endpoint contract	point difference	chemical-cluster 95%	target-jackknife 95%
all exact-relation endpoints (broad primary)	-0.008	-0.022 – 0.006	-0.056 – 0.030
human binding Ki/Kd	-0.040	-0.091 – 0.003	-0.133 – 0.029
single endpoint: Ki	-0.033	-0.069 – -0.005	–
single endpoint: Kd	+0.038	-0.008 – 0.080	–
single endpoint: IC50	-0.006	-0.022 – 0.011	–
single endpoint: EC50	+0.013	-0.013 – 0.042	–

c. Endpoint provenance of the 3,955 informative human binding Ki/Kd comparisons

Pair category	comparisons	fraction of all	fraction of endpoint-unambiguous
same endpoint (Ki–Ki: 1,358; Kd–Kd: 2,252)	3,610	91.3%	99.1%
mixed endpoint (Ki–Kd)	33	0.8%	0.9%
involves a cell pooled across Ki and Kd records	312	7.9%	–

Table S28: Post-hoc equivalence-margin decision curves for the residual-minus-absolute contrast. No margin was preregistered. The rule declares equivalence only when the conservative union of the two chemical-cluster 90% intervals lies strictly inside $(-\delta, +\delta)$. Because the decision curve is monotone in δ, its first passing grid value summarises the full 0.010–0.060 grid in 0.005 increments. These rows are sensitivity analyses only: passing a chosen margin does not show identical rules or transport beyond the observed within-ligand target comparisons.

Benchmark/stratum	point difference	conservative 90% interval	exact minimum symmetric margin	first passing grid margin	status
DAVIS	-0.009	-0.026 – 0.007	0.026	0.030	primary
DOCKSTRING-ChEMBL	-0.008	-0.019 – 0.004	0.019	0.020	primary
PKIS2	-0.003	-0.016 – 0.010	0.016	0.020	primary
DAVIS both uncensored	+0.044	0.010 – 0.081	0.081	>0.060	outcome-conditioned exploratory
DAVIS floor vs uncensored	-0.013	-0.034 – 0.006	0.034	0.035	outcome-conditioned exploratory
PKIS2 both active	+0.052	0.010 – 0.098	0.098	>0.060	outcome-conditioned exploratory

Table S29: Legacy sparse Docking-44–ChEMBL sensitivity (137 ligands, 38 targets, 691 observed cells and 2,522 non-tied pairs). This is retained only to show the effect of changing the docking matrix and matched chemical support; it is not the primary operational benchmark. The docking and external experimental priors are ligand-invariant target means fitted after excluding all evaluation ligands. The cohort outcome prior is scaffold-cross-fitted but benchmark-fitted and non-deployable. Target-centred and two-way-centred unscaled scores are exactly rank-identical because they differ only by a ligand-wise constant.

Representation	mean per-ligand accuracy	pair-weighted	top-1	evaluated ligands	conservative cluster 95% interval
ligand-invariant docking target-mean prior	0.525	0.585	0.277	137	0.456 – 0.591
ligand-invariant external experimental target-mean prior	0.525	0.565	0.270	137	0.465 – 0.584
scaffold-cross-fitted cohort outcome prior	0.592	0.644	0.409	137	0.529 – 0.654
absolute Vina	0.566	0.584	0.387	137	0.502 – 0.623
target-centred, unscaled	0.556	0.534	0.401	137	0.496 – 0.614
two-way-centred, unscaled	0.556	0.534	0.401	137	0.498 – 0.613
column-standardized Vina	0.546	0.513	0.372	137	0.485 – 0.605
scaled two-way residual	0.564	0.532	0.394	137	0.501 – 0.626

Table S30: Dense ligand-wise target-preference benchmarks on exact DOCKSTRING structure matches. DAVIS excludes exact pKd ties and pairs for which both affinities are at the pKd-5 floor; PKIS2 excludes experimental differences of 10 percentage points or less. Each estimate first averages pairwise concordance within ligand. Chemical-cluster intervals are the union of Murcko- and Butina-cluster 95% intervals. Target-centred/residual-SD scores apply target offsets and residual scales without ligand-row subtraction; scaled two-way residual scores apply row subtraction before target-specific scaling. Target-jackknife intervals refit all transformation parameters after each target deletion and are descriptive for the fixed 21-target panel.

a. Support and representation estimates

Panel	matched N	evaluated N	targets	within-ligand comparisons	target prior	absolute	column-standardized	target-centred	residual-SD scaled	scaled two-way residual
DAVIS	59	56	21	6,065	0.522	0.539	0.524	0.526	0.506	0.530
PKIS2	154	154	21	16,417	0.504	0.525	0.524	0.525	0.507	0.523

b. Baseline and transformation-path contrasts

Panel	absolute minus target prior	target-offset step minus absolute	residual-SD scaling minus unscaled	ligand-row correction under residual SD	net scaled residual minus absolute
DAVIS	+0.017 (-0.025 – 0.059)	-0.013 (-0.033 – 0.006)	-0.021 (-0.037 – -0.006)	+0.025 (0.011 – 0.039)	-0.009 (-0.030 – 0.010; target jackknife -0.081 – 0.061)
PKIS2	+0.021 (-0.006 – 0.048)	-0.001 (-0.016 – 0.015)	-0.017 (-0.028 – -0.006)	+0.015 (0.005 – 0.025)	-0.003 (-0.018 – 0.012; target jackknife -0.060 – 0.053)

Table S31: Exploratory outcome-conditioned strata in the dense target-preference benchmarks. DAVIS both-uncensored pairs compare two above-floor measurements, whereas floor-versus-uncensored pairs primarily test detection against the released pKd-5 floor. The primary PKIS2 both-active stratum requires both activities to be at least 65% and preserves the aggregate margin of more than 10 percentage points; the secondary row relaxes only that margin to exact non-ties. Stratum membership uses the experimental outcome, and all intervals are unadjusted for multiplicity; these rows generate hypotheses rather than replace the aggregate estimands.

Panel and stratum	ligands	within-ligand comparisons	target prior	absolute	scaled residual	residual minus absolute (cluster 95%)
DAVIS: both uncensored	49	1,757	0.476	0.459	0.503	+0.044 (0.004 – 0.089)
DAVIS: floor versus uncensored	56	4,308	0.535	0.547	0.534	-0.013 (-0.037 – 0.010)
PKIS2: both at least 65%, difference > 10	57	256	0.482	0.517	0.569	+0.052 (0.002 – 0.109)
PKIS2: both at least 65%, exact non-tie	66	591	0.448	0.514	0.562	+0.047 (0.004 – 0.094)

Table S32: Operational sensitivity to the evaluation-ligand row-mean target panel.

Quantity	estimate	Murcko 95% interval	Butina 95% interval	changed support
all-44 row-mean accuracy	0.564	–	–	137 ligands, 2,522 pairs
local-38 row-mean accuracy	0.567	–	–	137 ligands, 2,522 pairs
local-38 minus all-44	+0.0030	-0.0005–0.0072	-0.0006–0.0074	31 pair predictions in 19 ligands
local-38 minus absolute Vina	+0.0006	-0.0625–0.0658	-0.0606–0.0631	same fixed evaluation support

Table S33: Assay and coverage sensitivities for the legacy sparse Docking-44–ChEMBL benchmark. These rows do not replace the broad DOCKSTRING–ChEMBL primary benchmark. For the coverage rows, the scaffold-cross-fitted cohort outcome prior is fitted once on the full legacy cohort and the same predictions are evaluated on each nested stratum; it is a benchmark reference rather than a deployable rule.

Analysis	ligands	pairs	docking prior	cohort prior	absolute	column-standardized	scaled residual
all exact-relation endpoints	137	2,522	0.525	0.592	0.566	0.546	0.564
human binding Ki/Kd	107	2,224	0.514	0.608	0.565	0.554	0.579
same endpoint, equal endpoint weight	132	4,330	–	–	0.553	0.538	0.555
minimum 3 observed targets	93	2,478	0.590	0.636	0.587	0.504	0.530
minimum 5 observed targets	52	2,327	0.593	0.618	0.581	0.513	0.540
minimum 6 observed targets	40	2,207	0.584	0.621	0.587	0.519	0.536